Lattice Metamaterials: Theory

Lecture 3 – Topologies, energy absorption, design for impact

Dr. L. Lomazzi

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Roadmap

1. **Part A** – Mechanical metamaterials: a primer
2. **Part B** – Lattice topologies and the unit-cell idea
3. **Part C** – Beyond trusses: TPMS and Voronoi lattices
4. **Part D** – Energy absorption mechanisms
5. **Part E** – Design strategies for impact loading
6. **Part F** – Wrap-up and bridge to Lecture 4

What this afternoon adds

Morning sessions: physics of dynamic loads and deep-learning tools.
This afternoon: the **material class** that best exploits both – architected lattices.

L3 – What is a metamaterial?

Classical material. Its behavior comes from its *chemistry and microstructure*: bonds, grains, phases. Property range set by the periodic table.

Metamaterial. Its behavior comes from its *geometry at a scale well below the component*: repeating architected features (cells, struts, channels). Property range set by **what we design**.

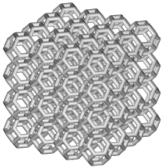
Three length scales matter.

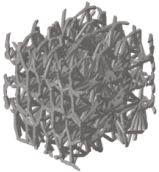
- *Base material* – the bulk from which we build (e.g. Ti6Al4V, polymer, stainless steel).
- *Unit cell* – the repeating architectural feature (mm scale, typically).
- *Component* – the full structural part (cm–m scale).

L3 – Mechanical metamaterials specifically

We focus on **mechanical** metamaterials: targeting stiffness, strength, and energy absorption.

- **Access to property combinations not available in bulk.**
  - Ultralight and stiff.
  - Tunable Poisson's ratio.
  - Fluid-like solids (pentamodes) and negative-stiffness media.
- **Tunability by geometry.** Change the architecture ⇒ change the properties.
- **Multi-functionality.** A single lattice can simultaneously support load, absorb energy, and channel heat or fluid.





L3 – Canonical examples of mechanical metamaterials

Auxetic materials.

- Negative Poisson's ratio: they get *thicker* transversely when stretched.
- Re-entrant honeycombs, rotating-squares, chiral geometries.
- Applications: impact protection, medical stents.

Pentamode metamaterials.

- Solid that behaves almost like a fluid – resists compression, not shear.
- Narrow-strut cubic diamond lattice.

Negative-stiffness composites.

- Locally unstable inclusions in a stable matrix.
- High damping, shock mitigation.

Ultralight lattices.

- Very small density with retained stiffness.
- The subclass most relevant to **this course**: energy absorption under impact.

L3 – Effective properties and homogenization

Why can we treat an architected material as "a material" at all?

If the unit-cell size a is much smaller than the component size L_{comp}, we can **homogenize**:

$$a \ll L_{\text{comp}} \quad \Rightarrow \quad \text{treat the lattice as a continuum with effective } E^\star,\ \nu^\star,\ \sigma_y^\star,\ \rho^\star.$$

The effective Young's modulus depends on the architecture, not only on the base material:

$$E^\star = E_s \cdot \mathcal{F}(\text{geometry},\ \bar{\rho})$$

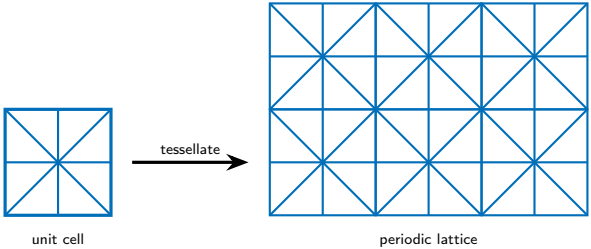
with E_s the solid-material modulus, $\bar{\rho} = \rho^\star/\rho_s$ the **relative density** (fraction of space occupied by material).

Design lever

The function $\mathcal{F}$ is our design knob. Two lattices with the *same base material and same mass* can have $E^\star$ differing by an order of magnitude, depending on their architecture.

L3 – The unit cell and the lattice

A lattice is obtained by periodically tessellating a **unit cell** across space.



Two knobs to turn.

- **Topology** – which struts / surfaces / walls appear in the unit cell. Discrete choice.
- **Relative density** $\bar{\rho}$ – how thick those features are. Continuous knob.

Three families we will meet

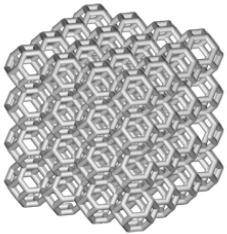
Truss lattices · **TPMS metamaterials** · **Voronoi lattices**

L3 – Truss lattices

Unit cell = a network of slender **struts** (beams) meeting at nodes.

Canonical 3D topologies.

- **Octet-truss** – regular octahedra sharing vertices with tetrahedra; very stiff for its mass.
- **Kelvin foam** – truncated octahedron; packs space uniformly.
- **Diamond** – tetrahedrally coordinated, characteristic "isotropic" response.



Each topology has a different **connectivity** Z (struts per node) and a different ratio of axial-to-bending deformation under load $\Rightarrow$ wildly different effective properties even at the same $\bar{\rho}$.

L3 – Stretching- vs bending-dominated lattices

The single most important classification for truss lattices is according to **Maxwell's criterion**:

$$M = b - 2n + 3 \quad (2\text{D}), \qquad M = b - 3n + 6 \quad (3\text{D}),$$

with b the number of struts and n the number of nodes.

- $M \geq 0$: **stretching-dominated**.
- $M < 0$: *bending-dominated*.

Consequence for effective stiffness.

$$\text{stretching-dominated:} \;\; E^\star \propto \bar{\rho}, \qquad \text{bending-dominated:} \;\; E^\star \propto \bar{\rho}^2.$$

L3 – Relative density and the Gibson–Ashby scaling

Relative density (the central design parameter):

$$\bar{\rho} = \frac{\rho^\star}{\rho_s} = \frac{\text{volume of struts}}{\text{volume of unit cell}}, \qquad \bar{\rho} \in (0, 1).$$

Gibson–Ashby scaling laws (empirical, validated across many lattices):

$$\frac{E^\star}{E_s} \approx C_1 \, \bar{\rho}^{\,n_E}, \qquad \frac{\sigma^\star_{\text{pl}}}{\sigma_{y,s}} \approx C_2 \, \bar{\rho}^{\,n_\sigma}$$

- $E^\star, \sigma^\star_{\text{pl}}$ – effective modulus and plateau stress; $E_s, \sigma_{y,s}$ – solid-material properties.
- $n_E \approx 1$ (stretch-dominated) or ≈ 2 (bending-dominated).
- $n_\sigma \approx 1.5$–2 depending on mode (plastic bending, buckling).

L3 – Anisotropy in lattices

A lattice's properties depend on **which direction you load it**, as the unit cell has a finite set of symmetries. Loading along a symmetry axis activates struts axially; loading off-axis bends them.

Consequences.

- A cubic lattice (BCC, FCC, octet) is *cubic anisotropic*: three independent elastic constants, not two.
- Peak stiffness along one axis can be 2–5$\times$ the off-axis stiffness.
- Under multi-axial loading, a "good" lattice in one direction may fail catastrophically in another.

Design implication

Real applications have **multi-directional loading**. Options: (i) choose a more isotropic topology (diamond, Kelvin), (ii) *orient* the lattice to align its strong axis with the dominant load.

L3 – TPMS: Triply Periodic Minimal Surfaces

A **minimal surface** has *zero mean curvature* everywhere – it minimizes area locally. A **TPMS** is a minimal surface that is periodic in all three spatial directions.

How to build a TPMS lattice.

- Define the surface implicitly: $\phi(\mathrm{x}) = 0$, with ϕ a trigonometric function.
- Thicken it into a solid shell, or fill one side of the surface with material.

Canonical TPMS:

- **Gyroid**: $\phi = \sin x \cos y + \sin y \cos z + \sin z \cos x$.
- **Schwarz P** (primitive): $\phi = \cos x + \cos y + \cos z$.

L3 – Why TPMS are attractive for impact

Geometric properties.

- **Smooth and curvature-continuous** – no sharp corners, hence no stress concentrations.
- **Self-supporting** in many orientations – easier to print.

Mechanical consequences.

- Lower peak stresses $\Rightarrow$ later failure initiation under dynamic load.
- More uniform deformation $\Rightarrow$ a longer, flatter plateau in stress–strain.
- Higher SEA (specific energy absorption) than most truss lattices at equal $\bar{\rho}$.

When TPMS wins

When the application prioritizes *energy absorption per unit mass* and the manufacturing process is AM. The gyroid is the current de-facto benchmark for impact protection.

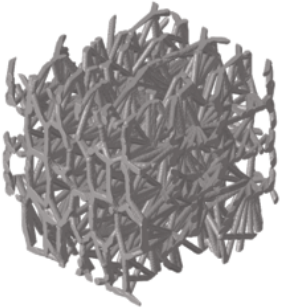
L3 – Voronoi lattices

Voronoi lattices are random lattices built by:

1. Scattering N seed points in the unit cell.
2. Computing the **Voronoi tessellation**.
3. Extract the edges of the cells as struts.

What this buys us.

- Mimics natural microstructures – bone trabeculae, open-cell foams.
- Progressive failure: no aligned rows of struts collapsing together.



L3 – Three families side by side

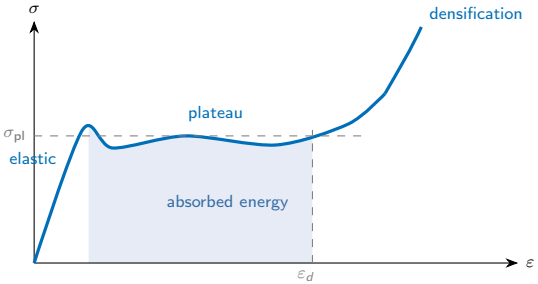
	Truss	**TPMS**	**Voronoi**
Representation	graph of struts	implicit surface $\phi = 0$	tessellation of seeds
Design variables	topology + thickness	type + thickness + period	seeds + thickness
Manufacturability	needs supports	often self-supporting	can have overhangs
Peak stress	high (at nodes)	low (smooth surface)	moderate
SEA	moderate (BCC/Kelvin)	high (gyroid)	high (progressive)
Design-space dim.	small–moderate	small–moderate	large (high-dim)

Rule of thumb

Truss → well understood, fast to optimize · *TPMS*→ best SEA / mass under AM · Voronoi → most flexible, needs DL to navigate.

L3 – The stress–strain signature of a crushable lattice

Characteristic compression response of a lattice:



- **Elastic regime** – governed by $E^\star$.
- **Plateau** – (near-)constant stress σ_{pl} while cells deform/collapse. This is where energy is absorbed.
- **Densification** at strain ε_d – cells have fully collapsed, stress ramps up.

L3 – The three key performance metrics

Key performance metrics for metamaterials for energy absorption:

1. **Plateau stress** (σ_{pl}) – the near-constant stress level during progressive collapse.
2. **Densification strain** (ε_d) – when the lattice "bottoms out" and stress increases exponentially.
3. **Specific energy absorption** (SEA) – work done per unit mass:

$$\text{SEA} = \frac{1}{\rho^\star} \int_0^{\varepsilon_d} \sigma(\varepsilon)\, d\varepsilon \approx \frac{\sigma_{\text{pl}}\, \varepsilon_d}{\rho^\star} \quad [\text{J/kg}].$$

- Primary metric for comparing crashworthy materials.
- A *good* lattice has high $\sigma_{\text{pl}}\, \varepsilon_d$ at low $\rho^\star$.

L3 – Deformation modes under compression

What actually happens inside the unit cell during the plateau:

Mechanism	**What happens**	**When / SEA impact**
Elastic buckling	Slender struts buckle before yielding; recoverable if elastic, otherwise triggers plastic collapse.	Low-density lattices of stiff materials.
Plastic hinge formation *(folding)*	Plastic hinges form at strut midspans / joints; cells fold progressively.	Bending-dominated lattices. **Long flat plateau ⇒ good SEA.**
Brittle fracture / crushing	Struts shatter rather than fold.	Ceramic or highly optimized bases. **Oscillating stress ⇒ bad SEA.**

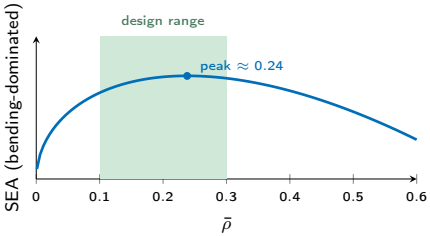
L3 – Scaling: σ_{pl} vs relative density

Gibson–Ashby plateau stress for ductile lattices:

$$\frac{\sigma^{\star}_{\mathrm{pl}}}{\sigma_{y,s}} \approx C\,\bar{\rho}^{\,3/2} \quad \text{(bending-dominated)}, \qquad \frac{\sigma^{\star}_{\mathrm{pl}}}{\sigma_{y,s}} \approx C\,\bar{\rho} \quad \text{(stretch-dominated)}.$$

Densification strain decreases with density:

$$\varepsilon_d \approx 1 - \alpha\,\bar{\rho}, \qquad \alpha \approx 1.4\text{–}2.0.$$



L3 – Strain-rate effects under impact

Everything above assumes quasi-static loading. Under impact ($\dot{\varepsilon} \gtrsim 10\,\mathrm{s}^{-1}$), three new effects kick in:

1. Material rate sensitivity. Yield stress $\sigma_{y,s}$ rises with $\dot{\varepsilon}$ (Johnson–Cook, Cowper–Symonds). Plateau stress scales up.

2. Inertial stabilization of struts. Dynamic loading delays buckling – struts carry more load before collapsing. Raises the plateau, sometimes substantially.

3. Shock / compaction wave. At high velocities, a compaction front propagates through the lattice. The effective plateau stress seen by the impactor is

$$\sigma_{\mathrm{dyn}} \approx \sigma_{\mathrm{pl}} + \frac{\rho^{\star} v^2}{\varepsilon_d} \quad \text{(Reid–Peng 1D shock model).}$$

Implication for design

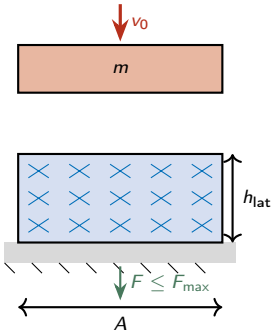
Quasi-static tests *under-predict* the plateau at high rates. Either test dynamically (Day 5), or include rate effects in the FE model *before* feeding data to the DL surrogate.

L3 – What "design for impact" actually means

Given. Impactor mass m, velocity v_0; protected object tolerates $F \leq F_{\max}$.

Three design constraints.

- Energy to absorb: $KE = \frac{1}{2} m v_0^2$.
- Force cap: $\sigma_{\text{pl}}^{\star} \leq F_{\max}/A$.
- Required stroke: $s \geq \dfrac{KE}{F_{\max}}$ (ideal plateau).



Design knobs.

- $\sigma_{\text{pl}}^{\star}$, $\varepsilon_d \Leftarrow \bar{\rho}$, topology, base material.
- Stroke $s \Leftarrow$ lattice thickness h_{lat}.
- Mass $m_{\text{lat}} = \rho^{\star} A\, h_{\text{lat}}$ – *minimize*.

L3 – Strategy 1: functional grading

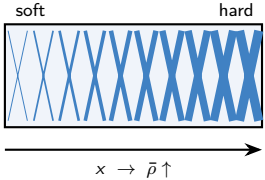
Idea. Vary $\bar{\rho}$ (or topology) continuously through the thickness:

$$\bar{\rho}(x) \;=\; \bar{\rho}_{\text{min}} + \left(\bar{\rho}_{\text{max}} - \bar{\rho}_{\text{min}}\right)\left(x/h_{\text{lat}}\right)^{n}.$$

Two design choices.

- *Soft face / hard back*: progressive deceleration, bounded peak force.
- *Hard face / soft back*: absorbs initial pulse, protects remaining energy.

Grading direction and exponent n are design variables.



DL-friendly

$\bar{\rho}(x)$ is a continuous *field* – ideal input for a CNN or neural operator.

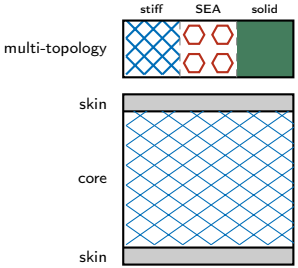
L3 – Strategy 2: multi-topology & hybrid structures

Multi-topology. Different regions use different cells:

- Octet (stretch) where stiffness matters.
- Kelvin (bending) where SEA matters.
- Smooth transitions to avoid stress concentrations.

Hybrid structures.

- *Sandwich*: solid skins + lattice core.
- *Hierarchical*: nested length scales of cells.



Design-space explosion

Huge number of possible configurations ⇒ classical optimization loses; **generative / inverse** DL wins.

L3 – Manufacturing constraints

AM reality check.

- Min. feature $\sim$ 0.2–0.5 mm (metal L-PBF).
- Overhangs $< 45^\circ$: need supports.
- Enclosed voids: trap powder.
- Surface roughness 10–50 μm: lowers effective properties.
- Residual stresses: distortion, microcracks.

Design discipline

Put manufacturability *in* the objective – not post-hoc.

L3 – Pitfalls

- Lower $\bar{\rho} \neq$ more SEA. Peak at *intermediate* $\bar{\rho}$.
- Quasi-static $\neq$ impact. Rate effects matter.
- Good lattice $\neq$ good in every direction. Anisotropy plays an important role.
- TPMS $\not>$ truss always. Depends on goal and AM quality.
- Good cell is not enough. Grading and orientation often dominate.

Trade-off

Stiffness, SEA, mass, manufacturability – pick three.

L3 – Summary & bridge

What we covered.

- Architecture $>$ composition.
- Three families: trusses, TPMS, Voronoi.
- Metrics: $\sigma^{\star}_{\mathrm{pl}}$, ε_d, SEA – all scale with $\bar{\rho}$.
- Design for impact: grading, multi-topology.

Next

Design space + target performance $\Rightarrow$ Lecture 4: navigate it with DL.

Questions?

Lecture 4 to follow: Metamaterials – Application